



VELOCITY OF DENSITY



Define the **velocity** of $\dot{\pi}_\beta : X \rightarrow \mathbb{R}$ as the instantaneous rate position



The velocity coincides with the **Fishers** core ends the direction and rate change in density

$$\dot{\pi}_{\beta}(x) = \frac{\mathrm{d}}{\mathrm{d}\beta} \log \pi_{\beta}(x)$$

$$\dot{\pi}_{\beta}(x) = \lim_{\Delta\beta \rightarrow 0} \frac{\Delta\pi_{\beta,\beta+\Delta\beta}(x)}{\Delta\beta}$$

$$= \frac{\mathrm{d}}{\mathrm{d}\beta} \log \pi_{\beta}(x)$$

$$\dot{\pi}_{\beta}(x) \coloneqq \lim_{\Delta\beta \rightarrow 0} \frac{\Delta\pi_{\beta, \beta + \Delta\beta}(x)}{\Delta\beta}$$



Proot

$$= \lim_{\Delta\beta \rightarrow 0} \frac{\pi_{\beta+\Delta\beta}(x) - \pi_{\beta}(x)}{\Delta\beta \pi_{\beta}(x)}$$

$$= \frac{1}{\pi_{\beta}(x)} \frac{\mathrm{d}\pi_{\beta}}{\mathrm{d}\beta}(x)$$

Note that for the linear path, the Fisher score constant velocity (up to an additive constant)

$$\dot{\pi}_{\beta}(x) = V(x) - \frac{\mathrm{d}}{\mathrm{d}\beta} A(\beta)$$

VELOCITY OF DENSITY

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- Define the **velocity** of $\dot{\pi}_\beta : \mathbb{X} \rightarrow \mathbb{R}$ as the instantaneous rate position

$$\dot{\pi}_\beta(x) := \lim_{\Delta\beta \rightarrow 0} \frac{\Delta\pi_{\beta, \beta+\Delta\beta}(x)}{\Delta\beta}$$

- The velocity coincides with the **Fisher score** encodes the direction and rate of change in density

$$\dot{\pi}_\beta(x) = \frac{d}{d\beta} \log \pi_\beta(x)$$

- Note that for the linear path, the Fisher score constant velocity (upto an additive constant)

$$\dot{\pi}_\beta(x) = V(x) - \frac{d}{d\beta} A(\beta)$$

- Proof:**

$$\dot{\pi}_\beta(x) = \lim_{\Delta\beta \rightarrow 0} \frac{\Delta\pi_{\beta, \beta+\Delta\beta}(x)}{\Delta\beta} = \lim_{\Delta\beta \rightarrow 0} \frac{\pi_{\beta+\Delta\beta}(x) - \pi_\beta(x)}{\Delta\beta \pi_\beta(x)} = \frac{1}{\pi_\beta(x)} \frac{d\pi_\beta}{d\beta}(x) = \frac{d}{d\beta} \log \pi_\beta(x)$$

VELOCITY OF EXPECTATIONS